**BUSINESS ANALYTICS**

Group Project Announcement

# Predicting Customer Order Value for an E-Commerce Platform

*An End-to-End Analytics Project: From Raw Data to a Monitored, Deployed Model*

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|---|---|
| **Course** | Business Analytics |
| **Project Type** | Group Project (4–5 students per team) |
| **Domain** | E-Commerce — Revenue & Customer Value |
| **Deployment Target** | Cloud free tier (e.g., Render, Hugging Face Spaces, Streamlit Community Cloud) |
| **Coding Requirement** | All code must be written by team members (no no-code/AutoML platforms) |

## 1. Project Overview

E-commerce platforms rely on accurate revenue expectations to plan inventory, cash flow, marketing spend, and personalized promotions. Knowing in advance roughly how much a customer's current cart or upcoming order is worth allows the business to tailor offers and forecast revenue with much greater precision than relying on historical averages alone. This project asks each team to act as the revenue-analytics function of an e-commerce platform and answer a concrete business question:

> ***"Given a customer's profile and current cart/session behavior, what is the expected value of this order, and which factors drive that value up or down?"***

Teams will build a regression model that predicts the monetary value of an order (or a customer's next-order value) from customer profile, browsing/cart behavior, and product-mix features. The model will be engineered, deployed as a working application, and monitored after deployment — mirroring the full lifecycle of an analytics product in industry.

## 2. Data Sources

Teams must combine the following two data sources:

- **Kaggle dataset (required base):** "Brazilian E-Commerce Public Dataset by Olist" — kaggle.com/datasets/olistbr/brazilian-ecommerce. This provides order-level transactions, product categories, payments, and customer identifiers.
- **Synthetic data (required extension):** Each team must generate additional behavioral data using Python (e.g., the Faker library plus custom business logic) — session duration,

number of items viewed/added to cart, discount/coupon usage, referral channel (organic/ads/email), and device type.

This combination ensures the project is not a simple "download-and-train" exercise: teams must justify the realism of the behavioral features they generate and document the simulation logic rigorously.

**Data Dictionary Requirement:** Every synthetically generated field must be documented in a data dictionary specifying column name, data type, unit, valid range, and the generation logic or business assumption used.

## 3. Project Structure (8 Modules)

The project is organized into eight modules, suitable for an 8-week timeline (adjustable to fit the course schedule).

| Module | Focus Area | Key Tasks |
|---|---|---|
| **1** | **Business Understanding & Data Generation** | • Define the business problem and key KPIs (prediction error tolerance, business value of accurate revenue estimates)<br>• Generate supplementary session/behavioral data in Python<br>• Produce a complete data dictionary |
| **2** | **Exploratory Data Analysis (EDA)** | • Analyze distributions, outliers, and correlations between customer/session attributes and order value<br>• Compare order value patterns across product category, referral channel, and customer segment<br>• Visualize relationships (e.g., order value vs. items in cart, vs. discount usage) and check for multicollinearity |
| **3** | **Data Cleaning** | • Handle missing values, duplicates, and inconsistent categorical labels (e.g., product category naming)<br>• Treat outliers (e.g., implausible order values, returned/cancelled orders that distort totals)<br>• Document all cleaning decisions with before/after comparisons |
| **4** | **Feature Engineering** | • Construct derived features (items per session, average item price, discount rate, customer recency/frequency)<br>• Encode categorical variables, scale numerical features, and address skewness in order value (e.g., log transform)<br>• Perform feature selection and check for multicollinearity (e.g., VIF) |
| **5** | **Model Development** | • Train and compare regressors: Linear/Ridge/Lasso Regression, Random Forest, XGBoost / LightGBM<br>• Evaluate with regression metrics: RMSE, MAE, MAPE, and $R^2$; validate with k-fold cross-validation |

| Module | Focus Area | Key Tasks |
|---|---|---|
| | | • Analyze residuals to check model assumptions and identify systematic under/over-prediction (e.g., for high-value orders) |
| **6** | **Model Deployment** | • Package the model as an API (FastAPI or Flask) that returns a predicted order value with a confidence range<br>• Build a demo interface (Streamlit) simulating a real-time order-value estimator for the checkout/marketing team<br>• Deploy to a cloud free tier (Render, Hugging Face Spaces, or equivalent) |
| **7** | **Model Monitoring** | • Track data drift by comparing incoming session/order data distributions to training data<br>• Monitor rolling prediction error (RMSE/MAE) over time as new orders complete<br>• Build a simple monitoring dashboard (e.g., using the open-source Evidently library) and define retraining triggers |
| **8** | **Final Report & Presentation** | • Translate findings into marketing and promotion-targeting recommendations based on key value drivers<br>• Discuss limitations and future improvements<br>• Present to the class with a live or recorded demo |

## 4. Deliverables

| Deliverable | Format |
|---|---|
| Source code (notebooks/scripts) | GitHub repository |
| Data dictionary | Excel or Markdown file |
| Deployed API + order-value estimator demo | Live link (Render / Hugging Face Spaces / Streamlit Cloud) |
| Monitoring dashboard or report | Notebook or Evidently report |
| Final report and slides | PDF and PPTX |

## 5. Suggested Team Roles (4–5 members)

| Role | Responsibilities |
|---|---|
| **Data Engineer** | Synthetic session/behavioral data generation, data dictionary, data cleaning |
| **Data Analyst** | Exploratory data analysis and visualization of order-value drivers |

| Role | Responsibilities |
|---|---|
| **ML Engineer — Modeling** | Feature engineering and regression model development |
| **ML Engineer — Evaluation** | Cross-validation, residual analysis, and error-metric interpretation |
| **MLOps Lead** | Deployment and monitoring (may be combined with another role for 4-person teams) |

## 6. Evaluation Focus

Grading will emphasize the rigor and clarity of each stage, rather than model accuracy alone:

- Quality and realism of synthetic behavioral data, supported by a complete data dictionary
- Depth of EDA and clarity of insights communicated through visualization
- Soundness of cleaning and feature engineering decisions, with clear documentation
- Correct use of cross-validation and appropriate regression metrics (RMSE, MAE, MAPE, $R^2$)
- A working, accessible deployment (live link required)
- A functioning monitoring component with defined drift/performance triggers
- Originality of code — all code must be authored by team members

## 7. Submission Logistics

Instructors should insert specific dates for each milestone below before distributing this announcement.

1. Team formation and topic confirmation — [insert date]
2. Module 1–3 checkpoint (data + EDA + cleaning) — [insert date]
3. Module 4–5 checkpoint (features + models) — [insert date]
4. Module 6–7 checkpoint (deployment + monitoring) — [insert date]
5. Final submission and presentation — [insert date]

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*Questions about scope, data sources, or deployment options should be directed to the course instructor.*